SHORT ANSWER QUESTIONS

UNIT – I

S No	Question	BTL	CO	POs / PSOs
1	What is the definition of an Information Retrieval (IR) system?	**1**	CO1	PO1, PO2, PO12
2	What are the primary objectives of an IR system?	**1**	CO1	PO1, PO2, PO12
3	Name three core functional components of an IR system	**1**	CO1	PO1, PO2, PO12
4	What is the role of indexing in IR systems?	**1**	CO1	PO1, PO2, PO12
5	How does query processing differ from matching in IR?	**1**	CO1	PO1, PO2, PO12
6	How does IR differ from traditional DBMS-based data retrieval?	**1**	CO1	PO1, PO2, PO12
7	What types of data do DBMS systems typically handle compared to IR systems?	**1**	CO1	PO1, PO2, PO12
8	What is structured retrieval in the context of IR?	**1**	CO1	PO1, PO2, PO12
9	Can relational DBMS support IR-like querying? Yes—explain briefly.	**2**	CO1	PO1, PO2, PO12
10	What unique advantage does IR offer over DBMS in handling unstructured text?	**1**	CO1	PO1, PO2, PO12
11	What is a digital library in relation to IR systems?	**1**	CO1	PO1, PO2, PO12
12	Name one protocol used for metadata harvesting in digital libraries (e.g., OAI-PMH).	**2**	CO1	PO1, PO2, PO12
13	How does full-text search work in digital library IR?	**1**	CO1	PO1, PO2, PO12
14	What role do metadata and cataloging play in digital libraries?	**1**	CO1	PO1, PO2, PO12
15	List one key challenge faced by IR systems in digital libraries.	**1**	CO1	PO1, PO2, PO12
16	What is a data warehouse?	**1**	CO1	PO1, PO2, PO12
17	How does a data warehouse differ from an IR system in focus and usage?	**1**	CO1	PO1, PO2, PO12
18	What does ETL stand for in a data warehouse context?	**1**	CO1	PO1, PO2, PO12
19	Explain the role of metadata in a data warehouse.	**2**	CO1	PO1, PO2, PO12
20	What is the main objective of integrating data from multiple sources into a warehouse?	**1**	CO1	PO1, PO2, PO12
21	What search query types do IR systems support? (e.g., Boolean, natural language)	**1**	CO1	PO1, PO2, PO12
22	Define fuzzy search in the IR context.	**2**	CO1	PO1, PO2, PO12
23	What is proximity search? Give a brief example.	**1**	CO1	PO1, PO2, PO12
24	What is concept/thesaurus expansion during query search?	**1**	CO1	PO1, PO2, PO12
25	How do probabilistic algorithms evaluate search results?	**1**	CO1	PO1, PO2, PO12

26	What is faceted search?	**1**	CO1	PO1, PO2, PO12
27	Give one example of drill-down filtering in IR.	**1**	CO1	PO1, PO2, PO12
28	How does browsing complement search in modern IR?	**1**	CO1	PO1, PO2, PO12
29	Name a use of faceted classification in digital library catalogs.	**2**	CO1	PO1, PO2, PO12
30	What is content-aware indexing in enterprise search?	**1**	CO1	PO1, PO2, PO12
31	What is relevance feedback in IR?	**1**	CO1	PO1, PO2, PO12
32	Define personalization in IR systems.	**1**	CO1	PO1, PO2, PO12
33	Name two multimedia types modern IR systems can handle. (e.g., images, audio)	**2**	CO1	PO1, PO2, PO12
34	Give one example of natural language processing in IR.	**1**	CO1	PO1, PO2, PO12
35	What IR component manages tokenization and stemming?	**1**	CO1	PO1, PO2, PO12
36	List two scenarios where IR and DBMS functionalities overlap.	**2**	CO1	PO1, PO2, PO12
37	Why is indexing scalable storage essential in IR?	**1**	CO1	PO1, PO2, PO12
38	What browsing/search capabilities would support a hybrid digital library–warehouse system?	**1**	CO1	PO1, PO2, PO12
39	How would fuzzy search help with OCR-imperfect scanned texts?	**2**	CO1	PO1, PO2, PO12
40	Name one emerging feature in IR (e.g., semantic linking, visualization).	**2**	CO1	PO1, PO2, PO12

SHORT ANSWER QUESTIONS

UNIT – II

S No	Question	BTL	CO	POs / PSOs
1	Define cataloging.	1	CO2	PO1, PO2, PO12
2	What is indexing?	1	CO2	PO1, PO2, PO12
3	Mention two objectives of cataloging.	1	CO2	PO1, PO2, PO12
4	What is automatic indexing?	1	CO2	PO1, PO2, PO12
5	Define stemming.	1	CO2	PO1, PO2, PO12
6	What is an inverted index?	1	CO2	PO1, PO2, PO12
7	What is a signature file?	1	CO2	PO1, PO2, PO12
8	State any two stemming algorithms.	1	CO2	PO1, PO2, PO12
9	What is an N-gram data structure?	1	CO2	PO1, PO2, PO12
10	What does PAT in PAT tree stand for?	1	CO2	PO1, PO2, PO12

11	Define a posting list.	1	CO2	PO1, PO2, PO12
12	What is a hyperlink?	1	CO2	PO1, PO2, PO12
13	What is term frequency (TF)?	1	CO2	PO1, PO2, PO12
14	Define information extraction (IE).	1	CO2	PO1, PO2, PO12
15	Name one hypertext-based algorithm.	1	CO2	PO1, PO2, PO12
16	Explain the purpose of cataloging in IR systems.	2	CO2	PO1, PO2, PO12
17	Demonstrate the indexing process in short.	2	CO2	PO1, PO2, PO12
18	Explain how does automatic indexing differ from manual indexing?	2	CO2	PO1, PO2, PO12
19	Classify the role of stemming in indexing.	2	CO2	PO1, PO2, PO12
20	Explain the structure of an inverted index.	2	CO2	PO1, PO2, PO12
21	Explain the function of the dictionary in inverted indexing?	2	CO2	PO1, PO2, PO12
22	Explain how a signature file works.	2	CO2	PO1, PO2, PO12
23	Explain how do N-grams help in information retrieval?	2	CO2	PO1, PO2, PO12
24	Show what is the function of a PAT tree?	2	CO2	PO1, PO2, PO12
25	Explain how hypertext structures support navigation.	2	CO2	PO1, PO2, PO12
26	Demonstrate how Page Rank uses hypertext links.	2	CO2	PO1, PO2, PO12
27	Demonstrate difference between stemming and lemmatization.	2	CO2	PO1, PO2, PO12
28	Extend the importance of information extraction in IR?	2	CO2	PO1, PO2, PO12
29	Illustrate how a bit vector is used in signature files.	2	CO2	PO1, PO2, PO12
30	Classify the significance of anchor text in hypertext systems?	4	CO2	PO1, PO2, PO12
31	Apply stemming to the word “connected” using Porter’s algorithm.	3	CO2	PO1, PO2, PO12
32	Build a simple inverted index for a document containing the words: "data", "index", "data", "search".	3	CO2	PO1, PO2, PO12
33	List bigrams for the word “catalog”.	3	CO2	PO1, PO2, PO12
34	Choose a simplified PAT tree for the words “cat”, “can”, “cap”.	3	CO2	PO1, PO2, PO12
35	Organize a basic signature for the keyword “indexing” using hashing.	3	CO2	PO1, PO2, PO12
36	Compare forward index and inverted index.	2	CO2	PO1, PO2, PO12
37	Apply NER to extract names from the sentence: “Google was founded by Larry Page and Sergey Brin.”	3	CO2	PO1, PO2, PO12
38	Utilize hyperlinks to connect three web pages in a hypertext structure.	3	CO2	PO1, PO2, PO12
39	Analyze the benefits of using hypertext data structures for web navigation.	4	CO2	PO1, PO2,

				PO12
40	Develop a mini-case study comparing inverted index and hypertext index in search engines.	**3**	CO2	PO1, PO2, PO12

SHORT ANSWER QUESTIONS

UNIT – III

S No	Question	BTL	CO	POs / PSOs
1	What is document clustering?	**1**	CO3	PO1, PO2, PO12
2	Define thesaurus generation in the context of information retrieval.	**1**	CO3	PO1, PO2, PO12
3	What is item clustering?	**2**	CO3	PO1, PO2, PO12
4	What does the hierarchy of clusters refer to?	**1**	CO3	PO1, PO2, PO12
5	Explain the purpose of document clustering in data analysis.	**1**	CO3	PO1, PO2, PO12
6	Relate thesaurus generation improve search results?	**1**	CO3	PO1, PO2, PO12
7	Demonstrate the process of item clustering.	**2**	CO3	PO1, PO2, PO12
8	Show hierarchy of clusters important in organizing information?	**1**	CO3	PO1, PO2, PO12
9	Apply document clustering to a set of news articles?	**1**	CO3	PO1, PO2, PO12
10	Build a scenario to use thesaurus generation for a search engine?	2	CO3	PO1, PO2, PO12
11	Select an example of item clustering in a retail context.	1	CO3	PO1, PO2, PO12
12	Identify implementation of, hierarchy of clusters in a library database?	1	CO3	PO1, PO2, PO12
13	Compare document clustering and item clustering.	2	CO3	PO1, PO2, PO12
14	Theme the effectiveness of thesaurus generation?	1	CO3	PO1, PO2, PO12
15	Analyze the benefits of using a hierarchy of clusters in data organization.	1	CO3	PO1, PO2, PO12
16	What challenges might arise when implementing document clustering?	2	CO3	PO1, PO2, PO12
17	Evaluate the effectiveness of different algorithms used for document clustering.	1	CO3	PO1, PO2, PO12
18	Assess the impact of thesaurus generation on user search behavior.	1	CO3	PO1, PO2, PO12
19	Measure the success of item clustering in a product recommendation system?	1	CO3	PO1, PO2, PO12
20	Design a framework for implementing a hierarchy of clusters in a digital library system.	2	CO3	PO1, PO2, PO12
21	What are the main classes of automatic indexing?	1	CO3	PO1, PO2, PO12
22	Compare statistical indexing from concept indexing?	1	CO3	PO1, PO2, PO12
23	Identify examples of natural language processing in automatic indexing?	2	CO3	PO1, PO2, PO12
24	List the advantages of using hypertext linkages in indexing?	1	CO3	PO1, PO2,

				PO12
25	Justify effectiveness of statistical indexing with natural language indexing?	1	CO3	PO1, PO2, PO12
26	Show a simple automatic indexing system that incorporates concept indexing.	2	CO3	PO1, PO2, PO12
27	What is the definition of automatic indexing?	1	CO3	PO1, PO2, PO12
28	Explain the role of natural language in automatic indexing.	1	CO3	PO1, PO2, PO12
29	Plan implementation of statistical indexing in a database?	2	CO3	PO1, PO2, PO12
30	Compare the efficiency of different classes of automatic indexing.	1	CO3	PO1, PO2, PO12
31	Assess the impact of hypertext linkages on user navigation in indexed content.	1	CO3	PO1, PO2, PO12
32	Propose a new method for integrating statistical and concept indexing.	2	CO3	PO1, PO2, PO12
33	List the types of automatic indexing.	1	CO3	PO1, PO2, PO12
34	Explain how concept indexing enhances search results.	1	CO3	PO1, PO2, PO12
35	Solve how natural language processing improves indexing accuracy?	2	CO3	PO1, PO2, PO12
36	Analyze the challenges arise when using hypertext linkages in indexing?	1	CO3	PO1, PO2, PO12
37	Critique the effectiveness of automatic indexing in digital libraries.	1	CO3	PO1, PO2, PO12
38	Develop a framework for evaluating the performance of indexing methods.	2	CO3	PO1, PO2, PO12
39	What is the purpose of automatic indexing?	**1**	CO3	PO1, PO2, PO12
40	Summarize the key features of statistical indexing.	**1**	CO3	PO1, PO2, PO12

SHORT ANSWER QUESTIONS

UNIT – IV

S No	Question	BTL	CO	POs / PSOs
1	What is a Boolean query?	1	CO4	PO1, PO2, PO12
2	Define AND, OR, and NOT in Boolean searches.	1	CO4	PO1, PO2, PO12
3	What role does stemming play in Boolean search?	2	CO4	PO1, PO2, PO12
4	What is phrase search in hypertext indexing?	1	CO4	PO1, PO2, PO12
5	What is the purpose of NEAR queries?	1	CO4	PO1, PO2, PO12
6	Why is query binding useful?	1	CO4	PO1, PO2, PO12
7	What is selective dissemination of information (SDI)?	2	CO4	PO1, PO2, PO12
8	How does relevance feedback refine a Boolean query?	1	CO4	PO1, PO2, PO12
9	What is a similarity measure in IR?	1	CO4	PO1, PO2,

				PO12
10	Give an example of a scalar similarity score.	2	CO4	PO1, PO2, PO12
11	How does cosine similarity work?	1	CO4	PO1, PO2, PO12
12	What does the Jaccard coefficient measure?	1	CO4	PO1, PO2, PO12
13	How is Dice similarity different from Jaccard?	2	CO4	PO1, PO2, PO12
14	Define TF-IDF weighting.	1	CO4	PO1, PO2, PO12
15	What is Okapi BM25?	1	CO4	PO1, PO2, PO12
16	What is the basic idea behind vector space modeling?	2	CO4	PO1, PO2, PO12
17	What is ranking in IR systems?	1	CO4	PO1, PO2, PO12
18	Why is proximity of query terms in a document important?	1	CO4	PO1, PO2, PO12
19	Define precision and recall.	1	CO4	PO1, PO2, PO12
20	What is the break-even point in IR evaluation?	2	CO4	PO1, PO2, PO12
21	Name a fuzzy-logic approach to ranking scores	1	CO4	PO1, PO2, PO12
22	What is explicit relevance feedback?	1	CO4	PO1, PO2, PO12
23	What is implicit feedback?	2	CO4	PO1, PO2, PO12
24	Define pseudo (blind) relevance feedback.	1	CO4	PO1, PO2, PO12
25	How does the Rocchio algorithm use feedback?	1	CO4	PO1, PO2, PO12
26	What is normalized discounted cumulative gain (nDCG)?	2	CO4	PO1, PO2, PO12
27	What is dwell time in implicit feedback?	1	CO4	PO1, PO2, PO12
28	How does SDI benefit users?	1	CO4	PO1, PO2, PO12
29	What is PageRank?	2	CO4	PO1, PO2, PO12
30	How does HITS differ from PageRank?	1	CO4	PO1, PO2, PO12
31	What is an inverted index	1	CO4	PO1, PO2, PO12
32	What is latent semantic indexing (LSI)?	2	CO4	PO1, PO2, PO12
33	What is the purpose of collaborative filtering?	1	CO4	PO1, PO2, PO12
34	Describe cross-lingual IR (CLIR)	1	CO4	PO1, PO2, PO12
35	Why visualize search results?	2	CO4	PO1, PO2, PO12
36	How might perception principles affect visualization design?	1	CO4	PO1, PO2, PO12
37	What is multi-reference-point visualization?	1	CO4	PO1, PO2, PO12

38	Name one technology used in IR visualization interfaces.	2	CO4	PO1, PO2, PO12
39	How does clustering aid information visualization?	1	CO4	PO1, PO2, PO12
40	Give one challenge in visualizing high-dimensional IR data	2	CO4	PO1, PO2, PO12

SHORT ANSWER QUESTIONS

UNIT – V

S No	Question	BTL	CO	POs / PSOs
1	What is the naive string search algorithm?	1	CO5	PO1, PO2, PO12
2	What's the time complexity of naive string search?	1	CO5	PO1, PO2, PO12
3	How does the Knuth–Morris–Pratt (KMP) algorithm improve on naive search?	2	CO5	PO1, PO2, PO12
4	What preprocessing structure does KMP use?	1	CO5	PO1, PO2, PO12
5	Define the Boyer–Moore algorithm's bad-character rule.	1	CO5	PO1, PO2, PO12
6	What is the good-suffix rule in Boyer–Moore?	1	CO5	PO1, PO2, PO12
7	Why is Boyer–Moore efficient for long patterns?	2	CO5	PO1, PO2, PO12
8	How does Rabin–Karp use hashing for pattern search?	1	CO5	PO1, PO2, PO12
9	What is the average-case complexity of Rabin–Karp?	1	CO5	PO1, PO2, PO12
10	What makes Aho–Corasick suited for multi-pattern search?	2	CO5	PO1, PO2, PO12
11	What is a trie, and how is it used in search systems?	1	CO5	PO1, PO2, PO12
12	Why might compressed tries benefit hardware-based search?	1	CO5	PO1, PO2, PO12
13	Describe a suffix tree or array's role in hardware querying.	2	CO5	PO1, PO2, PO12
14	What advantage do finite-state automata offer in hardware search?	1	CO5	PO1, PO2, PO12
15	How do hardware solutions parallelize string matching across patterns?	1	CO5	PO1, PO2, PO12
16	Why is memory usage critical in hardware text search acceleration?	2	CO5	PO1, PO2, PO12
17	How do specialized circuits accelerate regular-expression matching?	1	CO5	PO1, PO2, PO12
18	In what contexts is hardware search preferred over software implementations?	1	CO5	PO1, PO2, PO12
19	What does precision measure in IR evaluation?	1	CO5	PO1, PO2, PO12
20	What does recall quantify?	2	CO5	PO1, PO2, PO12
21	Define the F1 score	1	CO5	PO1, PO2, PO12
22	What is click-through rate (CTR) in search evaluation?	1	CO5	PO1, PO2, PO12

23	What does zero-result rate indicate?	2	CO5	PO1, PO2, PO12
24	Define session-success rate.	2	CO5	PO1, PO2, PO12
25	What does “queries per time” reveal about search performance?	1	CO5	PO1, PO2, PO12
26	Why is measuring dwell time useful?	1	CO5	PO1, PO2, PO12
27	What is an evaluation “ranking curve”?	2	CO5	PO1, PO2, PO12
28	How is top-k precision different from overall precision?	1	CO5	PO1, PO2, PO12
29	Why is F1 harmonic mean preferred over arithmetic mean here?	2	CO5	PO1, PO2, PO12
30	What is TREC, and what purpose does it serve?	1	CO5	PO1, PO2, PO12
31	What constitutes a TREC "run" in evaluation terms?	1	CO5	PO1, PO2, PO12
32	How are relevance judgments (qrels) used in TREC?	2	CO5	PO1, PO2, PO12
33	What is pooled evaluation in TREC?	1	CO5	PO1, PO2, PO12
34	Name a common TREC track (e.g., ad hoc, web).	1	CO5	PO1, PO2, PO12
35	Explain how TREC compares precision at various recall levels.	2	CO5	PO1, PO2, PO12
36	How is nDCG used in TREC-like evaluations?	1	CO5	PO1, PO2, PO12
37	What role do fairness-aware tracks play in TREC?	1	CO5	PO1, PO2, PO12
38	What is multi-aspect evaluation (e.g., relevance + credibility)?	2	CO5	PO1, PO2, PO12
39	How can zero-result rate affect TREC analysis?	1	CO5	PO1, PO2, PO12
40	Why benchmark systems against TREC pilots before deployment?	2	CO5	PO1, PO2, PO12

LONG ANSWER QUESTIONS

UNIT – I

S No	Question	BTL	CO	POs / PSOs
1	Define an Information Retrieval (IR) system and Explain its primary objectives in terms of indexing, search, and user satisfaction. How does this differ from general data storage systems?	2	CO1	PO1, PO2, PO12
2	Demonstrate the functional architecture of an IR system: explain components such as query formulation, matching algorithms, indexing, and user interface. How do they interplay to meet user needs?	3	CO1	PO1, PO3, PO12
3	Contrast IR systems with Database Management Systems (DBMS) in terms of data models, query languages, and typical applications. What are the experiential differences for users and developers?	2	CO1	PO1, PO2, PO12
4	Explain the integration between IR and DBMS systems, such as retrieval-augmented databases (e.g., Inquire DBMS). What benefits and challenges arise in terms of performance and representation?	3	CO1	PO1, PO3, PO12
5	Explain the role of IR systems in digital libraries, highlighting their function in full-text retrieval, metadata access, and interoperability using protocols like OAI-PMH.	2	CO1	PO1, PO2, PO12

6	Compare digital libraries, virtual libraries, and hybrid libraries, focusing on collection scope, metadata design, search modalities, and user experiences	2	CO1	PO1, PO3, PO12
7	Examine the challenges specific to digital libraries, including authentication, copyright, interface design, and long-term preservation. What IR strategies can help address these issues?	4	CO1	PO1, PO2, PO12
8	Define a data warehouse and contrast its structure and goals with those of IR systems. Explain the role of historical, structured data in decision-support settings.	2	CO1	PO1, PO3, PO12
9	Illustrate the concept of data marts within a warehouse architecture: how do they facilitate analytical querying compared to broader IR-based search on metadata?	2	CO1	PO1, PO2, PO12
10	Analyze the synergy between IR and data mining (KDD) in a data warehouse context. How can IR enhance pattern discovery, and vice versa?	4	CO1	PO1, PO3, PO12
11	Explain in Detail about the variety of search capabilities supported by modern IR systems, including Boolean queries, proximity, fuzzy search, term weighting, and natural language understanding.	2	CO1	PO1, PO2, PO12
12	Outline term weighting in search queries: how does assigning weights to query terms (e.g., "data mining(0.9) vs warehouses(0.3)") influence relevance ranking?	2	CO1	PO1, PO2, PO12
13	Examine natural language querying in IR: contrasting keyword-based modes with full NL parsing—what are the pros, cons, and user impacts?	4	CO1	PO1, PO2, PO12
14	Explain browsing capabilities such as faceted navigation and drill-down filtering. How do they complement textual search in enhancing discovery?	2	CO1	PO1, PO2, PO12
15	Identify miscellaneous IR features—recommendation engines, cross-database search, multimedia retrieval, and geospatial queries—and analyze their roles in richer IR experiences.	3	CO1	PO1, PO2, PO12
16	Explain in detail interactive visualizations (e.g., result-space graphs, statistical overviews) in IR tools and digital libraries. What perceptual principles justify their use?	3	CO1	PO1, PO2, PO12
17	Demonstrate the roles of IR, DBMS, digital libraries, and data warehouses: identify use cases where each excels, and scenarios where integrating them adds value.	2	CO1	PO1, PO3, PO12
18	Distinguish trade-offs in IR and DBMS hybrid systems: what tensions emerge between structure (DBMS) and relevance matching (IR)? Provide architectural suggestions.	3	CO1	PO1, PO2, PO12
19	Identify future evolution: as digital data grows, how must IR evolve in digital libraries and data warehouses—in terms of scalability, AI integration, and multimodality?	2	CO1	PO1, PO2, PO12
20	Construct an advanced IR system that supports search, browse, visualization, and analytics across structured and unstructured repositories. Describe architecture, indexing strategy, UI features, and evaluation metrics.	3	CO1	PO1, PO2, PO12

LONG ANSWER QUESTIONS

UNIT – II

S No	Question	BTL	CO	POs / PSOs
1	Explain the objectives of cataloging and indexing in digital information systems with suitable examples.	2	CO2	PO1, PO2, PO12
2	Demonstrate the step-by-step process of indexing and explain how it supports efficient retrieval of documents.	1	CO2	PO1, PO3, PO12
3	Explain the concept of automatic indexing. How is it better than manual indexing in large-scale systems?	2	CO2	PO1, PO2, PO12
4	show how stemming improves the performance of information retrieval systems. Illustrate with examples.	1	CO2	PO1, PO3, PO12
5	Explain the structure and working of an inverted file indexing system.	4	CO2	PO1, PO2, PO12
6	Compare and contrast signature file structure and inverted file structure with respect to storage and performance.	2	CO2	PO1, PO3, PO12
7	Illustrate the working of the N-gram data structure using a sample input text.	2	CO2	PO1, PO2, PO12

8	Explain the need for information extraction in digital libraries and web search.	2	CO2	PO1, PO3, PO12
9	Interpret the major components of a PAT (PATRICIA) tree? Explain its role in pattern matching.	2	CO2	PO1, PO2, PO12
10	Illustrate the hypertext data structure and explain how it differs from traditional linear data structures used in indexing.	4	CO2	PO1, PO3, PO12
11	Construct an inverted index for a document collection containing three documents. Show terms, document IDs, and positions.	2	CO2	PO1, PO2, PO12
12	Apply the Porter stemming algorithm to a list of words and show each step of transformation.	3	CO2	PO1, PO2, PO12
13	Build a basic N-gram index for the sentence: "Information retrieval is useful". Use bigrams and trigrams.	3	CO2	PO1, PO2, PO12
14	Build a PAT tree with an example of three strings. Show the insertion process.	2	CO2	PO1, PO2, PO12
15	Construct a signature file for a document collection using a simple hashing function. Discuss its effectiveness.	4	CO2	PO1, PO2, PO12
16	Apply information extraction techniques on a sample paragraph and identify named entities, locations, and dates.	2	CO2	PO1, PO2, PO12
17	Build a simple hyperlink structure between five web pages and explain how users would navigate the content.	2	CO2	PO1, PO3, PO12
18	Analyze the limitations of traditional indexing methods and explain how hypertext indexing addresses these limitations.	2	CO2	PO1, PO3, PO12
19	Compare automatic indexing and manual indexing in terms of scalability, accuracy, and cost-effectiveness.	3	CO2	PO1, PO2, PO12
20	Perform a case study on the hypertext data structure: a. Explain its structure (nodes, anchors, links). b. Analyze how it is used in modern search engines (e.g., PageRank). c. Discuss its advantages, challenges, and real-world relevance.	4	CO2	PO1, PO2, PO12

LONG ANSWER QUESTIONS

UNIT – III

S No	Question	BTL	CO	POs / PSOs
1	What are the main classes of automatic indexing?	1	CO3	PO1, PO2, PO12
2	Explain how statistical indexing differ from natural language indexing?	2	CO3	PO1, PO3, PO12
3	Apply concept indexing uses in information retrieval?	3	CO3	PO1, PO2, PO12
4	List the advantages and disadvantages of using hypertext linkages in automatic indexing?	4	CO3	PO1, PO3, PO12
5	Estimate effectiveness of statistical indexing compared to natural language indexing in terms of accuracy?	4	CO3	PO1, PO2, PO12
6	Design a new indexing system that incorporates elements from statistical and concept indexing. What features would it include?	2	CO3	PO1, PO3, PO12
7	List the key characteristics of natural language indexing	2	CO3	PO1, PO2, PO12
8	Explain the role of hypertext linkages in enhancing the searchability of indexed content.	3	CO3	PO1, PO3, PO12
9	How would you implement statistical indexing in a digital library system?	3	CO3	PO1, PO2, PO12
10	Assess the impact of automatic indexing on the efficiency of information retrieval systems.	4	CO3	PO1, PO3, PO12
11	What is document clustering, and how does it differ from term clustering?	2	CO3	PO1, PO2, PO12

12	Demonstrate thesaurus generation in the context of information retrieval. What are its primary purposes?	2	CO3	PO1, PO2, PO12
13	Explain the concept of item clustering. How does it contribute to the organization of information?	3	CO3	PO1, PO2, PO12
14	Explain the hierarchy of clusters. Why is it important in the context of document and term clustering?	3	CO3	PO1, PO2, PO12
15	Apply thesaurus generation be to improve search engine results. Identify specific examples.	2	CO3	PO1, PO2, PO12
16	Identify how item clustering can be utilized in a real-world application, such as e-commerce or academic research	4	CO3	PO1, PO2, PO12
17	Analyze the advantages and disadvantages of using hierarchical clustering methods compared to flat clustering methods in document clustering.	2	CO3	PO1, PO3, PO12
18	Compare and contrast different approaches to thesaurus generation. What factors should be considered when choosing an approach?	3	CO3	PO1, PO3, PO12
19	Evaluate the effectiveness of various clustering algorithms in the context of document clustering. Which algorithms do you find most effective, and why?	2	CO3	PO1, PO2, PO12
20	Design a new framework for item clustering that incorporates elements of thesaurus generation. What innovative features would you include, and how would they enhance the clustering process?	2	CO3	PO1, PO2, PO12

LONG ANSWER QUESTIONS

UNIT – IV

S No	Question	BTL	CO	POs / PSOs
1	Explain how weighted Boolean search systems extend classical Boolean logic. Compare their advantages and limitations in ranking document relevance compared to pure Boolean and probabilistic models.	2	CO4	PO1, PO2, PO12
2	Illustrate cosine similarity, Jaccard coefficient, and Dice similarity for document ranking. In which retrieval contexts is each measure most effective?	2	CO4	PO1, PO3, PO12
3	Compare TF–IDF and Okapi BM25 in terms of term weighting, document length normalization, and real-world performance. Provide examples of retrieval scenarios where one is preferable to the other.	2	CO4	PO1, PO2, PO12
4	Demonstrate the theoretical foundations of probabilistic IR models versus vector space models. How do their approaches to relevance differ, and what implications does this have for system design?	2	CO4	PO1, PO3, PO12
5	Explain and compare explicit feedback, implicit feedback, and pseudo (blind) relevance feedback. How do they improve retrieval effectiveness, and what are their practical challenges?	2	CO4	PO1, PO2, PO12
6	Explain the Rocchio algorithm in detail, including its mathematical formulation (weights α, β, γ). Under what conditions does it enhance retrieval, and when can it lead to query drift?	2	CO4	PO1, PO3, PO12
7	How does query expansion integrate with selective dissemination of information (SDI)? Present a complete user flow from initial search, feedback, to updated results in an SDI system.	2	CO4	PO1, PO2, PO12
8	Compare PageRank and HITS. Explain how they exploit web graph structures, their computation methods, and suitability for hypertext retrieval.	2	CO4	PO1, PO3, PO12
9	Explain how inverted indexes are constructed and structured. Discuss index compression techniques and their impact on search performance and scalability.	2	CO4	PO1, PO2, PO12
10	Identify the principles behind LSI. How does it handle synonymy and polysemy, and what are its computational complexities and limitations?	3	CO4	PO1, PO3, PO12
11	Illustrate how human cognitive and perceptual principles (e.g. Gestalt laws, pre-attentive processing) inform effective information visualization design in IR systems.	2	CO4	PO1, PO2, PO12
12	Analyze current technologies (e.g., t-SNE, force-directed graphs, heatmaps) used for visualizing search result spaces. When should each be applied?	4	CO4	PO1, PO2, PO12
13	Explain multi-reference-point visualization for IR. How does it help users navigate high-dimensional document spaces? Provide real-world interface examples.	2	CO4	PO1, PO2, PO12

14	Explain how clustering algorithms (e.g., hierarchical, k-means) aid visualization and retrieval. What are the trade-offs in cluster granularity and interpretability?	2	CO4	PO1, PO2, PO12
15	Examine performance metrics like precision, recall, break-even point, nDCG. How do they inform ranking algorithm tuning and system evaluation?	3	CO4	PO1, PO2, PO12
16	Demonstrate how implicit signals (such as dwell time, click-through) are captured and utilized in IR systems. What biases might these signals introduce?	2	CO4	PO1, PO2, PO12
17	Explain pseudo-relevance feedback (PRF), including tf-idf term selection and query expansion. Under what conditions does PRF succeed or fail?	2	CO4	PO1, PO3, PO12
18	Distinguish iterative relevance feedback in conversational or voice-based IR settings. How does it differ from batch relevance feedback?	3	CO4	PO1, PO3, PO12
19	Compare neural ranking models (deep or shallow) with classical models like BM25 or vector space. Discuss data requirements, interpretability, and development trends.	2	CO4	PO1, PO2, PO12
20	Outline challenges in visualizing high-dimensional IR data. What techniques help users explore clusters, semantic relationships, and outliers effectively?	2	CO4	PO1, PO2, PO12

LONG ANSWER QUESTIONS

UNIT – V

S No	Question	BTL	CO	POs / PSOs
1	Compare and contrast the Knuth–Morris–Pratt (KMP), Boyer–Moore, and Rabin–Karp algorithms in terms of time complexity, preprocessing steps, and typical use cases. Reference real-world scenarios where each algorithm outperforms the others.	**2**	CO5	PO1, PO2, PO12
2	Demonstrate the underlying principles of the Aho–Corasick algorithm, especially its construction of a finite-state automaton with failure links. Explain why its time complexity remains linear regardless of dictionary size.	**2**	CO5	PO1, PO3, PO12
3	How does Boyer–Moore's two-shift heuristics (bad-character and good-suffix rules) enhance search efficiency compared to naive string search? Provide examples of how each rule skips text.	1	CO5	PO1, PO2, PO12
4	Explain in Detail the rolling hash mechanism in Rabin–Karp, including how rolling updates work and their advantages/drawbacks compared to full string comparison	2	CO5	PO1, PO3, PO12
5	Construct a large-scale search system: analyze trade-offs among KMP, Boyer–Moore, Rabin–Karp, and Aho–Corasick for static vs. dynamic pattern sets and single vs. multiple queries.	3	CO5	PO1, PO2, PO12
6	Explain how hardware-accelerated text search systems—such as FPGA-implemented finite automata or compressed tries—enhance throughput in network intrusion detection or DNA sequence analysis. Provide architecture-level insights.	2	CO5	PO1, PO3, PO12
7	Examine compressed trie structures in hardware text search: how do they balance memory efficiency and search speed? What are the engineering considerations for on-chip implementations?	4	CO5	PO1, PO2, PO12
8	Illustrate parallel search architectures, such as bit-parallelism and pipelined matching—how do these designs exploit hardware-level parallelism for regex and pattern matching?	3	CO5	PO1, PO3, PO12
9	Distinguish hardware vs. software search approaches: in which domains (e.g., cybersecurity, genomics, large IR systems) is hardware acceleration justified? Include latency, power, and scaling considerations.	3	CO5	PO1, PO2, PO12
10	Construct a hybrid software-hardware search pipeline, explaining how document preprocessing and query handling are split between CPU and ASIC/FPGA, including bottleneck identification.	3	CO5	PO1, PO3, PO12
11	Explain the history of IR evaluation metrics, from precision and recall to advanced measures like MAP and nDCG. Explain why single-figure summary metrics gained popularity over curve-based evaluations.	2	CO5	PO1, PO2, PO12
12	How do user-centered metrics—like click-through rate, zero-result rate, session-success rate, and query volume—complement classical IR metrics? Explain biases and usability interpretation.	2	CO5	PO1, PO2, PO12
13	Compare and contrast MAP, R-precision, bpref, RBP, and graded relevance measures (e.g., nDCG) when qrel sets are incomplete or graded. Cite their	2	CO5	PO1, PO2, PO12

	performance in partial-judgment environments.			
14	Explain pooled evaluation in TREC, detailing the mechanism of selecting top-k documents per topic for relevance judgments. Discuss limitations such as unjudged relevant documents.	2	CO5	PO1, PO2, PO12
15	Summarize linear utility metrics (e.g., T10U, T11U) used in TREC filtering tracks. Explain their formal definition, scaling, and the role of utility parameters.	2	CO5	PO1, PO2, PO12
16	Demonstrate the construction of a TREC "run", covering query topics, retrieval output formatting, and submission pipelines. How do runs facilitate fair cross-system comparison?	2	CO5	PO1, PO2, PO12
17	Explain 11-point interpolated precision–recall curves and Mean Average Precision (MAP). How are they computed, and why did TREC shift from one to the other?	2	CO5	PO1, PO3, PO12
18	Identify the robustness of TREC collections when new neural models are evaluated. Can collections from the 1990s still reliably assess modern systems? Cite recent findings	3	CO5	PO1, PO3, PO12
19	What is graded relevance in TREC and NTCIR contexts, and how are evaluation metrics like nDCG and NSR adapted for graded judgments?	1	CO5	PO1, PO2, PO12
20	Illustrate the evolution of IR evaluation paradigms based on TREC history, from early binary relevance and recall/precision to multi-aspect, graded, and utility-aware evaluation used today, referencing multi-aspect metrics research.	2	CO5	PO1, PO2, PO12